

AI/ML Model Benchmark & Matrix Analysis Report

1. Overview of ML Models (Excluding Row-Level Isolation Forest)

The DataSentinel ML Detection Engine operates at the pipeline telemetry level rather than the row level. The models evaluated are divided into three domains: VOLUME, DISTRIBUTION, and OPERATIONAL. Unlike the row-level Isolation Forest which performed poorly (F1=0.10) on raw claim rows, these models evaluate aggregated pipeline execution metrics.

2. Metrics Used for Evaluation

The models were evaluated using the following metrics on an adversarial test split:

- Precision: Proportion of flagged anomalies that were true anomalies.
- Recall: Proportion of actual anomalies that were correctly flagged.
- F1 Score: Harmonic mean of Precision and Recall (Primary selection metric).
- FPR (False Positive Rate): Proportion of normal runs incorrectly flagged.
- Latency (ms): Time taken to evaluate a single pipeline run.

3. Matrix Analysis & Scores

Domain	Model	Precision	Recall	F1	FPR	Latency (ms)
VOLUME	MAD	0.25	1.00	0.40	0.1250	0.00
VOLUME	EWMA	0.33	1.00	0.50	0.0833	0.00
VOLUME	CUSUM	1.00	1.00	1.00	0.0000	1.01
DISTRIBUTION	Wasserstein_Proxy	0.04	1.00	0.08	1.0000	1.00
DISTRIBUTION	KS_Proxy	0.50	1.00	0.67	0.0417	1.71
DISTRIBUTION	PSI_Proxy	0.50	1.00	0.67	0.0417	0.00
OPERATIONAL	IsolationForest	0.33	0.50	0.40	0.0870	0.00
OPERATIONAL	LOF	0.50	0.50	0.50	0.0435	0.00
OPERATIONAL	OneClassSVM	0.22	1.00	0.36	0.3043	15.62

4. Model Selection & Superiority Analysis

VOLUME Domain: CUSUM vs Others

CUSUM (Cumulative Sum Control Chart) drastically outperformed MAD and EWMA, achieving a perfect F1 score (1.00) and 0.0000 FPR. It is superior because it accumulates small deviations over time, making it highly sensitive to sustained volume shifts rather than reacting to single-point volatility like MAD.

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DISTRIBUTION Domain: KS_Proxy / PSI_Proxy vs Wasserstein

KS_Proxy and PSI_Proxy both achieved an F1 score of 0.67 with an FPR of 0.0417, whereas Wasserstein completely failed with an FPR of 1.0000 (F1=0.08). KS_Proxy and PSI_Proxy are better because they are robust, rank-based (KS) or bin-based (PSI) metrics that are less sensitive to massive extreme outliers, whereas Wasserstein distance is heavily distorted by extreme values (e.g., healthcare billing spikes).

OPERATIONAL Domain: LOF vs Pipeline Isolation Forest vs OneClassSVM

Local Outlier Factor (LOF) achieved the best F1 score (0.50) and the lowest FPR (0.0435). Pipeline-level Isolation Forest achieved lower precision (0.33) than LOF (0.50). LOF is superior here because pipeline operational metrics (like processing duration) can have complex local densities. An execution might be 'normal' for a heavy batch but 'anomalous' for a small batch, which LOF captures well by evaluating local neighborhood density rather than global partitioning like Isolation Forest.